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Permeability of Salt-Rock Samples: Insights into Hydrogen Transport

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Abstract

This study explored hydrogen permeability in four salt-rock samples from the Lotsberg Salt Formation and its caprock using a modified pressure pulse decay (PPD) system. Experiments were conducted under a constant confining pressure of 28.6 MPa, with hydrogen pressures ranging from 1 MPa to 7 MPa. Our findings show that for anhydrite and marlstone samples, permeability measurements correlated with the reciprocal average pressure due to the Klinkenberg effect, which was corrected to determine intrinsic permeabilities of $7.22 \times 10^{-22} \text{ m}^2$ for S1, $9.21 \times 10^{-21} \text{ m}^2$ for S3, and $2.31 \times 10^{-21} \text{ m}^2$ for S4. The sample from Ernestina caprock, primarily composed of 95% anhydrite with minor dolomite, shows lowest permeability. However, halite salt rock S2 exhibited deviations from the expected linear relationship at lower pressures, indicating enhancing slip and Knudsen flow effects due to its smaller pore sizes. This sample's behavior highlights the influence of nano-scale structural characteristics on flow phenomena, necessitating advanced flow model for analysis. By integrating the correction of transition flow into our analysis for S2, the intrinsic permeability is determined to be $17.3 \times 10^{-21} \text{ m}^2$.

Introduction

The advancement of renewable energy technologies is challenged by the need for effective greenhouse gas management and reliable renewable energy storage (Anekwe et al., 2025). Geological storage systems, such as depleted reservoirs and salt caverns, offer promising solutions for subsurface energy and greenhouse gas containment (Peng et al., 2025). Hydrogen is expected to become increasingly important in the global energy landscape (Younus et al., 2025). Underground hydrogen storage, particularly in salt caverns, has key advantages including high storage pressures, reduced environmental impact, enhanced safety and long-term stability (Wallace et al., 2021). Salt formations are impermeable, which results in an expectedly low fluid leakage rate, thereby providing reliable containment for hydrogen storage (Zill et al., 2021). Since most H_2 production projects are expected to commence in the near future, the technology of hydrogen storage in salt caverns can be rapidly scaled up considering the massive salt formations in Alberta, Canada, including the Lotsberg, Cold Lake, Prairie Evaporite, and Hubbard Evaporite salts, with an estimated areal extent of 1.2 million km^2 (Grobe, 1999; Hauck, 2021). However, the implementation of hydrogen storage faces significant technical uncertainties.

While Canadian underground formations have been extensively utilized for natural gas, hydrocarbons, and waste storage, hydrogen (H_2) storage remains largely unexplored due to significant uncertainties (Ali and Guo, 2021; Lemieux et al., 2019). A primary concern for hydrogen storage in salt caverns is potential hydrogen leakage; however, the governing mechanisms controlling H_2 transport through salt rocks remain poorly understood, with limited studies on storage characteristics. Salt rocks are generally considered impermeable and non-porous due to their isotropic stress conditions and tightly closed grain boundaries (Caglayan et al., 2020; Yuan et al., 2024b). Nonetheless, frequent hydrogen injection and withdrawal cycles, necessitated by its low volumetric energy density (Lankof and Tarkowski, 2020), may induce cyclic thermal and mechanical stresses that alter the petrophysical and geomechanical properties of salt formations (Urai et al., 2008; Zill et al., 2021), particularly those containing impurities. These stress and temperature variations could result in dynamic changes in salt-rock properties (Ghanbarzadeh et al., 2015).

Hydrogen migration through salt rocks is influenced by factors such as pore-network connectivity, mineralogical heterogeneity, and operational conditions (Dias et al., 2023; Leuger and Zapf, 2012). Accurate measurement of salt-rock permeability is therefore essential to evaluate the long-term integrity of hydrogen storage, particularly in heterogeneous salt deposits with interbedded impurity layers. The pressure pulse decay (PPD) method has been widely employed for determining salt-rock permeability under hydrostatic or triaxial loading conditions (Peach and Spiers, 1996; Popp et al., 2001). Previous investigations report measurable but extremely low permeabilities for salt rocks, typically between 10^{-16} to 10^{-21} m² (Cosenza et al., 1999; Zhang et al., 2020). Salt-rock permeability measured using nitrogen (N_2) increases with higher impurity content, exhibiting values between 2.00×10^{-19} m² and 8.64×10^{-16} m² (Zhang et al., 2020). However, gas permeabilities measured with helium or nitrogen, rather than hydrogen, may lead to permeability overestimation due to the slippage effect, even though measurements commonly employ mean gas pressures above 1.5 MPa.

Gas transport in salt rocks primarily occurs along grain boundaries or interconnected microcracks, rather than through the halite crystals themselves (Baar, 2013; Grgic et al., 2022; Yuan et al., 2024b). However, some studies have demonstrated that under appropriate temperature and pressure conditions, cracks and fractures can undergo self-healing (Chen et al., 2016; Zhu and Arson, 2015). From a microscopic perspective, this self-healing process is associated with the recrystallization of rock salt crystals. Recrystallization involves the movement of existing grain boundaries and the development of new, high-angle grain boundaries. This mechanism of grain boundary healing closely resembles the process of repairing intragranular cracks (Chan et al., 1998). Consequently, the closure or healing of cracks may restore permeability. Gas flow through porous media can occur in various regimes, including Darcy flow, slippage flow, transition flow, and free molecular flow, determined by the Knudsen number, which is the ratio of the gas mean free path to the pore throat radius (Xie et al., 2024; Ziarani and Aguilera, 2012). In salt rock, where pore throats range from micrometers to nanometers, gas flow primarily occurs in the Darcy and slippage flow regions (Cyran et al., 2023; Mao et al., 2023). Smaller pore sizes and lower gas pressures increase the slippage effect. In the slippage flow regime, the apparent permeability, as measured by Darcy's law, is higher than the intrinsic permeability (Jones, 1972; Shell Development Co. and Klinkenberg, 2012). Thus, when examining hydrogen flow through salt rocks, it is crucial to account for the gas slippage effect, or Klinkenberg effect, on permeability (Wu, 2016). There is limited research on the permeability characteristics of impure salt rock, cement, and cemented salt rock, as well as the impact of hydrogen slippage on permeability. Therefore, investigating these areas is crucial to offer dependable guidance for ensuring the pressure integrity of caverns used for hydrogen storage.

In this study, a modified pressure pulse decay (PPD) system was used to measure the porosity and H_2 permeability of salt-rock samples from the caprock, salt, and marlstone layers of the Lotsberg Salt Formation in Saskatchewan, Alberta. Before testing, samples were characterized using X-ray diffraction (XRD) to

determine mineral composition. Permeability measurements were conducted at varying H_2 pressures, with values calculated from transient pressure responses.

Materials

We selected salt rocks from the Lotsberg Formation to measure their permeability to hydrogen. The Lotsberg Salt Formation (LSF), located within Alberta's Industrial Heartland, is one of the major salt deposits of the Devonian Elk Point Group in the Western Canada Sedimentary Basin (WCSB) (Grobe, 1999; Hauck, 2020; Zivar et al., 2025). The LSF is composed mainly of halite, along with evaporite minerals such as dolomite, calcite, anhydrite, and gypsum, and is subdivided into Upper and Lower units by a distinct shale interval (Grobe, 1999). Salt thickness varies geographically, with the Upper and Lower Lotsberg formations reaching maximum thicknesses of approximately 165 m and 75 m, respectively (Grobe, 1999; Hauck, 2021). In the Fort Saskatchewan (FS) area, the Upper LSF has an average thickness of 47 m and hosts around 80% of Alberta's over 150 active caverns (Hauck, 2020). These caverns, constructed since the 1950s at depths ranging from 750 m to more than 2100 m, are primarily used for natural gas and petrochemical storage (Hauck, 2021; Royan, 1984; Toboła and Kukiałka, 2020). The extensive cavern utilization and favorable geological characteristics highlight LSF's strong potential for large-scale hydrogen storage development.

Different samples were collected from the salt layers, marlstone interbeds, and top anhydrite layer of the upper LSF in FS area. The upper LSF is composed of halite deposits and marlstone interlayers, and is topped by the Ernestina Lake Limestone (Yuan et al., 2024b). Salt caverns in this area are typically constructed within the upper LSF, extending through the different units. Salt samples from Salt units primarily consist of halite with minor dispersed impurities, whereas marlstone interbeds contain carbonate base interspersed with embedded halite crystals (Yuan et al., 2024b; Zivar et al., 2025). Although salt rocks are typically impermeable due to their crystalline structure (Caglayan et al., 2020; Zill et al., 2021), the presence of impurities and interbedded layers could potentially enhance hydrogen diffusion. Therefore, samples were selected from the caprock, Upper Salt, and Upper Marlstone layers (1860–1925 m) for mineralogical characterization and hydrogen permeability tests. Sample details, including IDs, depths, and descriptions, are presented in Table 1.

Table 1—The information of salt rock samples selected to conduct experiments.

Sample ID	Depth (m)	Description
S1	1867	Caprock: Limestone with anhydrite composition
S2	1897	Upper Salt: Halite Salt with inner crystal boundaries and impurities
S3	1903	Upper Marl: Marlstone sample with carbonate composition
S4	1911	Upper Marl: Marlstone sample with halite crystals

Methodology

This section presents the salt-rock characterization methods, and the modified pressure-pulse setup and technique used to measure H_2 permeability of salt rocks. The diffusion results are related to the salt-rock plugs' crystal boundaries and mineralogy.

Salt-Rock Characterization

X-Ray Diffraction Characterization. The sample mineral composition was analyzed using X-ray diffraction (XRD). Samples were first ground to particle sizes below 2 μm . A Rigaku Ultima IV diffractometer with copper radiation ($\lambda = 1.54 \text{ \AA}$) at 40 kV and 44 mA was employed for the analyses. Measurements were conducted at room temperature with a scanning range from 5° to $90^\circ 2\theta$, a step size of $0.02^\circ 2\theta$, and a scan speed of $2^\circ 2\theta$ per minute. A spinning sample holder was not used. Each measurement

utilized 1–2 grams of rock powder. However, due to the inherent heterogeneity of salt rocks, these small samples might not fully represent the overall mineralogical composition.

Hydrogen Permeability Measurements

Core Plugs Drilling. Core plugs were drilled from full-diameter cores using a 1-inch diamond core bit. Figure 1 shows core plugs drilled from three selected samples: the caprock, the Lotsberg salt layer, and the marlstone interbed, respectively. Sample S1, primarily composed of anhydrite, exhibits fewer impurities and crystal boundaries than the other samples. These plugs were subsequently used for hydrogen permeability tests. The porosity and permeability values of these plugs will be analyzed to investigate the hydrogen transport mechanisms within the salt rock formations.



Figure 1—The salt plugs (1-inch diameter) cored from the full-diameter samples: (a) S1 from the Ernestina Formation; (b) S2 from the LSF salt zone; (c) S3 from the LSF marlstone zone; and (d) S4 from the LSF marlstone zone.

Modified Pressure Pulse System. Figure 2 illustrates the schematic of this experimental apparatus, comprising a core holder connected to two pressurized hydrogen vessels and a pulse-free pump for applying confining pressure. Core plugs were placed inside the holder with the upstream and downstream chambers set to initial pressures, P_u and P_d , respectively. The test initiates by establishing a pressure difference across the plug, allowing hydrogen to permeate until equilibrium between the chambers is reached. Permeability was determined from the slope of pressure differences recorded over time. Experiments were performed at pore pressures between 1 and 7 MPa, under a constant effective stress of 28.6 MPa. The initial pore-pressure difference was kept below 0.5 MPa to minimize variations in gas properties. More details about the experimental method and data analysis can be found in our previously published papers (Yuan et al., 2025, 2024a).

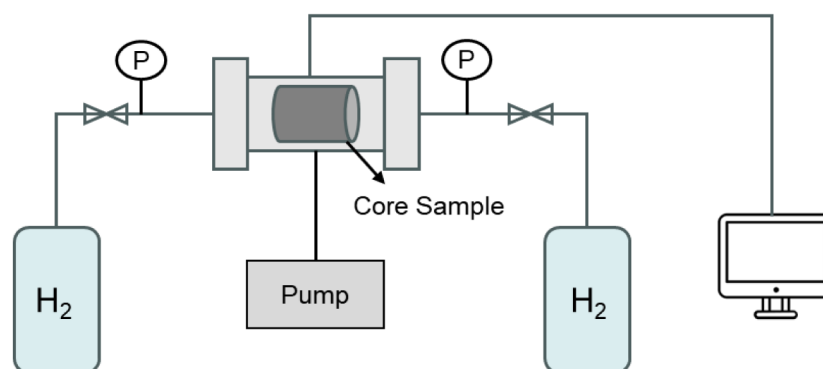


Figure 2—The modified pressure pulse decay system to measure salt-rock permeability.

Permeability Modelling

We used a numerical model using COMSOL Multiphysics for measuring the permeability of salt-rock samples. More details can be found in our published materials (Najafimarghmaleki et al., 2025; Yuan et al., 2025).

Numerical Modelling. The flow of H₂ through salt rock can be expressed using the following equation, which is based on the mass conservation equation:

$$\frac{\partial(\rho\phi)}{\partial t} = \frac{\partial}{\partial x} \left(\frac{k\rho}{\mu} \frac{\partial P}{\partial x} \right) \quad \text{Eq. 1}$$

Here ρ is the fluid density (kg/m³), ϕ is the porosity of the medium, t is time (s), x represents length (m), μ is gas viscosity (Pa.s), P is pressure (Pa), and k is apparent permeability (m²).

Eq. 1 can be expressed as follows after applying the chain rule and expanding the flux term:

$$\phi \frac{\partial \rho}{\partial P} \frac{\partial P}{\partial t} = \frac{\rho k}{\mu} \frac{\partial^2 P}{\partial x^2} + \frac{\partial \left(\frac{\rho k}{\mu} \right)}{\partial x} \frac{\partial P}{\partial x} \quad \text{Eq. 2}$$

In this study, we measured the permeability under room pressure and temperature. Therefore, the density and viscosity of hydrogen at 25 °C (experimental temperature in our study) can be obtained from the published literature (Assael et al., 1986; Yusibani et al., 2011):

The initial conditions can be expressed as follows:

$$p(x, 0) = \begin{cases} p_u(0), & x = 0 \\ p_d(0), & 0 < x \leq L \end{cases} \quad \text{Eq. 3}$$

Here $p_u(0)$ is the initial pressure in the upstream end, and $p_d(0)$ is the initial pressure in the entire system before the step change is applied.

The boundary conditions can be formulated as follows:

$$V_u \frac{\partial(\rho)}{\partial P} \frac{\partial P}{\partial t} \Big|_{x=0} = - \frac{\rho k A}{\mu} \frac{\partial P}{\partial x} \Big|_{x=0} \quad \text{Eq. 4}$$

$$V_d \frac{\partial(\rho)}{\partial P} \frac{\partial P}{\partial t} \Big|_{x=L} = \frac{\rho k A}{\mu} \frac{\partial P}{\partial x} \Big|_{x=L} \quad \text{Eq. 5}$$

Here A (m²) is the cross-sectional area of the core, and V_u (m³) and V_d (m³) are the upstream and downstream chamber volumes, respectively.

COMSOL Multiphysics was then used to solve the governing equation with the boundary conditions. The simulation was conducted to determine the permeability and porosity values that best match the experimental pressure profiles observed in the upstream and downstream vessels with the numerical predictions.

Results and Discussions

Salt-Rock Mineralogy

Four salt-rock samples were analyzed for permeability. Table 2 summarizes their mineral compositions at various depths, representing three zones: the caprock, the Lotsberg salt zone, and the marlstone interbed. Sample S1, from the caprock, is primarily composed of anhydrite, with minor amounts of quartz and dolomite. Sample S2, from the salt zone, consists mainly of halite with trace carbonate impurities such as dolomite, calcite, and nahcolite. Our previous work has shown that these carbonates are typically distributed along halite crystal boundaries (Yuan et al., 2022). Samples S3 and S4, from the marlstone zone, are composed of a halite-carbonate matrix, with carbonates interbedded with clays. Additional minerals identified in these samples include quartz, muscovite, and feldspar. Figure 3 presents the XRD patterns of all four samples.

Table 2—The XRD results of the LSF samples (in wt.%).

Depth (m)	Zone	Sample ID	Halite	Quartz	Clays/ Muscovite	Anhydrite	Feldspar	Carbonates			Other Minerals (lime, leucite, ankerite etc)
								Dolomite	Calcite	Other carbonates	
1867	Caprock	S1	0	2	-	96	-	2	-	-	-
1897	Upper Salt	S2	88	-	1	-	-	4	2	5	-
1903	Middle Marls	S3	44	11	-	-	-	18	23	-	4
1911	Upper Marls	S4	32	9	11	-	13	35	-	-	13

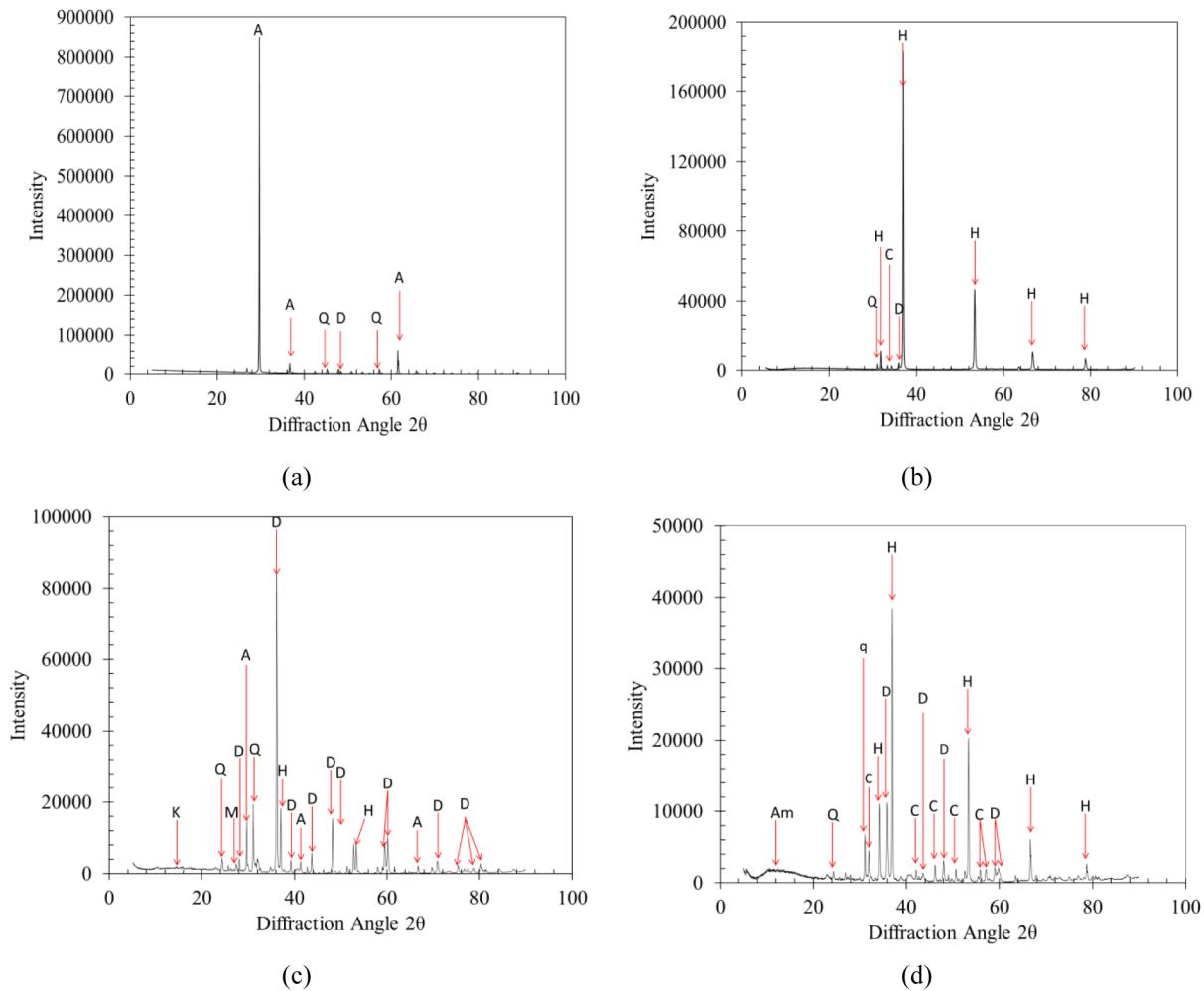


Figure 3—The example XRD patterns of (a) S1; (b) S2; (c) S3; (d) S4. 'A' represents anhydrite, 'H' represents halite, 'D' represents dolomite, 'C' represents calcite, 'Q' represents quartz, 'M' represents muscovite, 'K' represents kaolinite, 'Am' represents amorphous material.

Upstream and Downstream Pressure Profiles

Figure 4 illustrates the measured upstream and downstream pressure profiles during H_2 diffusion experiments conducted on salt-rock samples under a confining pressure of 28.6 MPa. Each subfigure presents the response of different salt-rock samples to varying upstream pressures. Figure 4a displays the pressure profiles for sample S1 located above the Lotsberg Formation within the caprock layer. A series of pressure decay tests were performed on this sample, with initial upstream pressures ranging from 1.59 MPa

to 6.81 MPa and downstream pressures adjusting from 1.18 MPa to 6.38 MPa. While the upstream pressure remained relatively constant, the downstream pressure gradually increased, eventually equilibrating with the upstream pressure. This is attributed to the larger volume of the upstream vessel relative to the pore volume and the downstream chamber volume. Figure 4b presents the pressure profiles for sample S2, primarily composed of halite. The results indicate that at lower pressures (< 4 MPa), the H_2 diffusion rate is faster compared to that observed under higher pressures, leading to quicker equilibration between downstream and upstream pressures. Figure 4c and Figure 4d show the pressure profiles for samples S3 and S4 from the marlstone layer, where the initial upstream pressures were incrementally varied from less than 1 MPa to 7 MPa in different diffusion tests. Despite slight declines, the upstream pressure remained largely stable across these tests. As H_2 diffuses through the sample, the downstream pressure increases until it reaches equilibrium with the upstream pressure. In contrast to sample S2, samples S3 and S4 exhibit a quicker H_2 diffusion rate and a shorter time to equilibrium as the pressures increase. Throughout these experiments, the rate of increase in downstream pressure was slowest for sample S1, indicating a lower permeability relative to the other tested samples. This differential pressure response can be numerically analyzed to get the permeability of salt rocks.

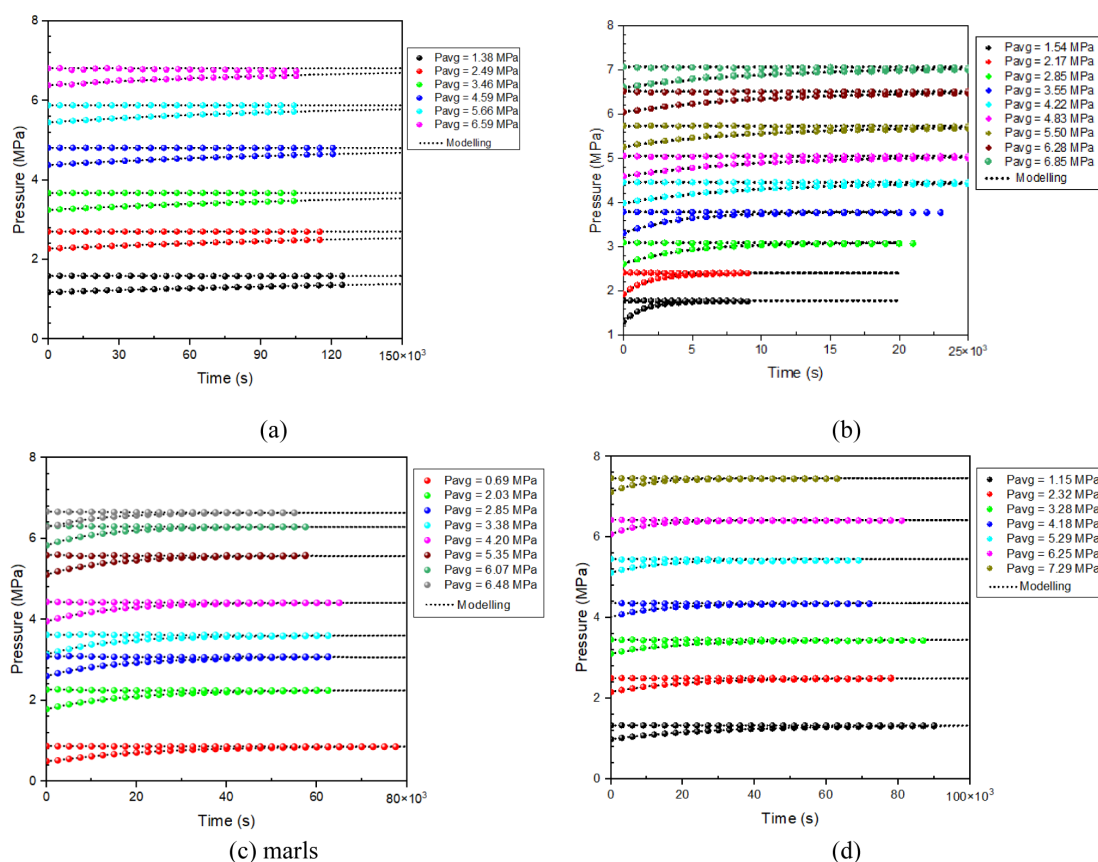


Figure 4—The measured and modelled upstream and downstream pressure during H_2 diffusion through the salt rock samples when setting the upstreaming pressure at different values: (a) S1; (b) S2; (c) S3; and (d) S4.

The Determination of Permeability Values of Salt-Rock Plugs

In this part, we analyzed the apparent permeability and also instinct permeability of the salt-rock plugs.

Klinkenberg's Correction. According to the PPD method, the permeability of salt-rock plugs can be determined at various pressures. All experimental data were analyzed through numerical modeling to fit the pressure data and subsequently determine the apparent permeability. Our model utilizes downstream pressure data to estimate the permeability and porosity of the salt-rock plugs, rather than the relatively

constant upstream pressure data. This method, which primarily relies on the downstream pressure transducer, reduces measurement errors. Additionally, it eliminates the need to wait for the system to reach final equilibrium state, thereby reducing test duration. Figure 5 illustrates the measured permeability of each plug versus the reciprocal average pressure for the four salt-rock samples. As shown in Figure 5a, c and d, the pressure data points demonstrate a linear relationship for S1, S3 and S4. The y-intercept represents the absolute H_2 permeability of the plugs, and the slope of each line corresponds to the Klinkenberg constant (b), reflecting the degree of slippage. This indicates that H_2 transport in these samples adheres to slippage flow. Klinkenberg's correction remains applicable, and the slip effect follows a first-order correction. As illustrated in the figures, the measured permeability decreases with increased pore pressure, which can be explained by the reduced slippage effect when pore pressure increases. The intrinsic permeability of S1, S3, and S4 is $7.22 \times 10^{-22} m^2$, $9.21 \times 10^{-21} m^2$ and $2.31 \times 10^{-21} m^2$ respectively. The calculated b values for S1, S3, and S4 are 4.19 MPa, 1.87 MPa, and 4.63 MPa, respectively. A higher b for sample 2 indicates a greater degree of slippage effect during H_2 flow.

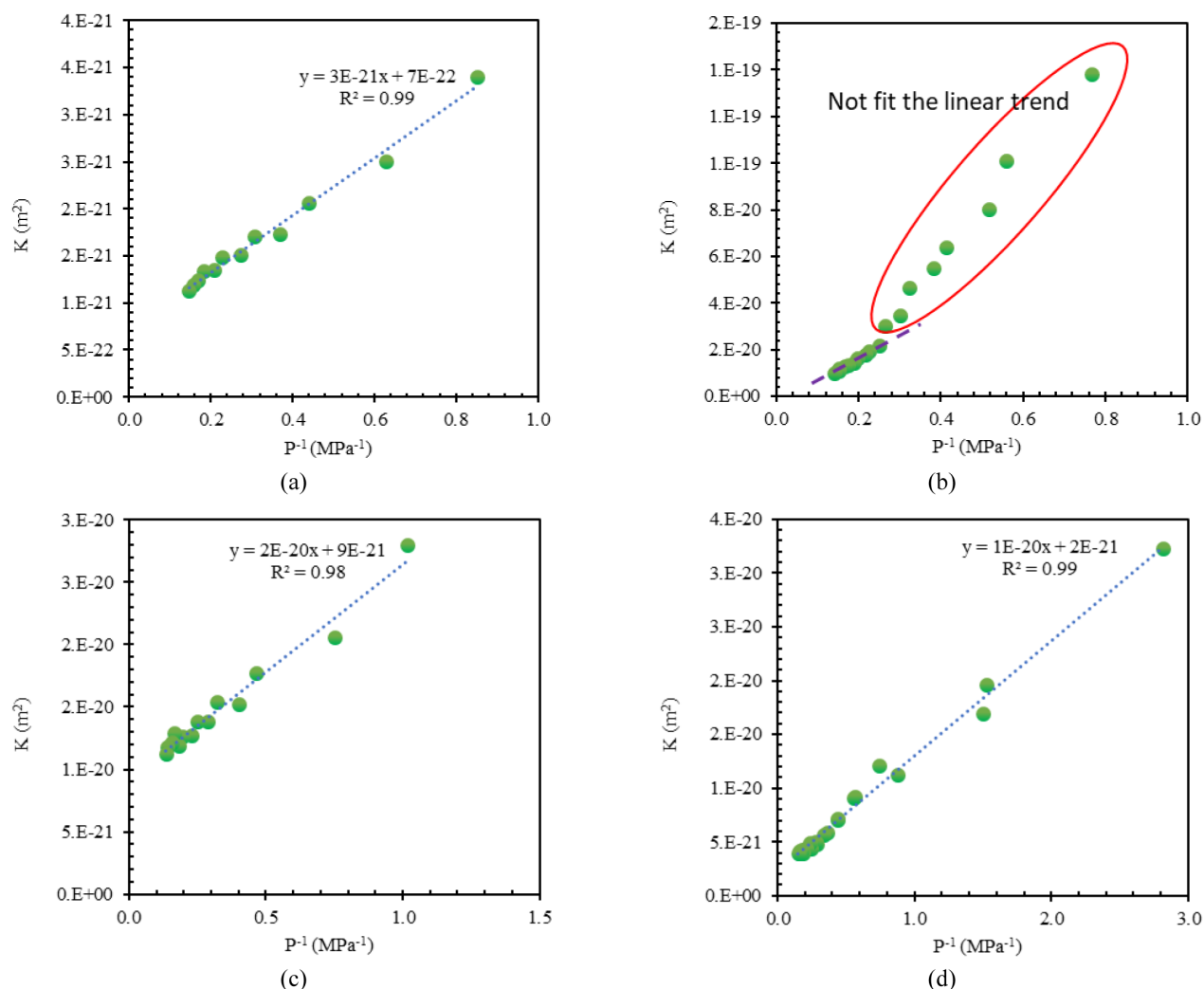


Figure 5—The determined permeability versus reciprocal of pressure for the experiments conducted on (a) S1; (b) S2; (c) S3; and (d) S4.

Transition Flow in Crystallized Halite. In the case of sample S2, as observed in Figure 5b, when the pressure is high (P^{-1} ranges from 0.14 - $0.25 MPa^{-1}$), the measured permeability maintains a linear relationship with the reciprocal of pressure. However, as the pressure decreases (corresponding to an increase in P^{-1}), deviations from this linear trend suggest potential transitions to enhanced slip effects in transition flow or Knudsen

flow. Unlike S2, for the other three samples (S1, S3, S4), lower average pressures resulted in longer times for downstream pressure to equilibrate with upstream pressure. But for S2, when the average pressure drops below 3.55 MPa, the H_2 diffusion rate increases, contrary to the behavior observed in the other samples. This anomaly indicates that instead of the Klinkenberg correction, a more advanced flow model considering transitional flow is required. Though grain boundaries exist in S2, the main component is crystallized halite. The hydrogen transport through crystallized halite may undergo transition flow, which is attributed to the extremely small pore sizes in crystallized halite matrix. The following equation is utilized to calculate gas permeability across the entire flow regime (Civan, 2010; Peng et al., 2021):

$$k = k_{\infty} (1 + \alpha(Kn)Kn) \left(1 + \frac{4Kn}{1 + Kn} \right) \quad \text{Eq. 6}$$

Here, $\alpha(Kn)$ is the rarefaction coefficient, which is a function of the Knudsen number, Kn (Civan, 2023), Kn is the Knudsen number, and $\left(1 + \frac{4Kn}{1 + Kn} \right)$ is a slip correction factor, which adjusts permeability based on Knudsen effects. This equation is similar to the Klinkenberg permeability correction but goes further by incorporating Knudsen diffusion effects, making it applicable in nano-porous materials and ultra-low permeability samples. Kn is given by the following equation:

$$Kn = \frac{\lambda}{d} \quad \text{Eq. 7}$$

Where λ is the gas mean free path and d is pore diameter.

Based on previous study, pore diameters of halite salt typically range from 2 and 8 nm (Song et al., 2024). Therefore, we use 5 nm as the value for modelling its permeability. For S2, when pore pressure decreases, the Knudsen number increases, shifting behavior more towards Knudsen flow. This phenomenon explains the observed pressure profile in Figure 5, where at lower pressures, hydrogen diffusion is accelerated. This acceleration is due to the increased mean free path of H_2 at low pressures, compounded by the small pore sizes in crystallized halite salt, leading to higher Knudsen numbers ($Kn > 1$). Consequently, H_2 molecules interact more frequently with pore walls than with other H_2 molecules, reducing the frequency of intermolecular collisions and thereby increasing diffusion rates. This explains why, for S2, lower pore pressures result in higher diffusion rates than those observed at higher pressures. By integrating considerations of transition flow and Knudsen flow into our analysis for S2, the porosity and true intrinsic permeability k_{∞} of it are determined to be 1.5% and $17.3 \times 10^{-21} \text{ m}^2$ respectively. Subsequently, using the intrinsic permeability k_{∞} for S2 and considering the effect of flow regime, we modeled the apparent permeability versus pore pressure as shown in Figure 6.

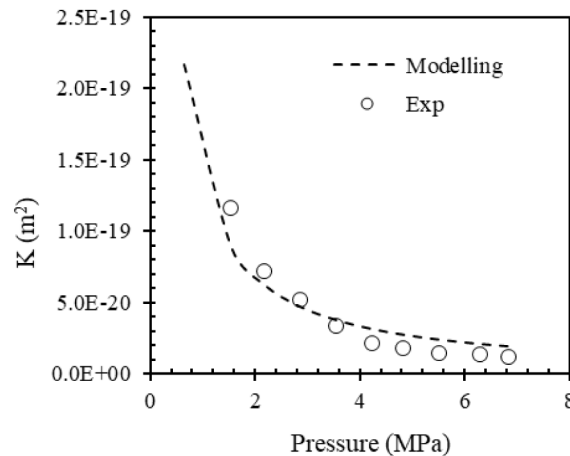


Figure 6—The measured and modelled apparent permeability versus pressure during H_2 diffusion through S2 when considering the effect of transitional flow.

Table 3 presents the measured porosity and intrinsic H₂ permeability for the tested samples. All samples exhibit ultra-low permeability, confirming the strong sealing potential of salt rocks, including the interbedded marlstone and caprock layer. Among them, sample S1 (caprock) shows the lowest permeability, suggesting that the Ernestina caprock, composed of anhydrite and carbonates, is an effective caprock for H₂ storage in salt caverns. In contrast, sample S2, from the halite-rich salt layer, shows the highest permeability. This may due to the developed grain boundaries and visible microfractures in S2, which may contribute to enhanced H₂ transport pathways. Samples S3 and S4, from the marlstone zone, exhibit intermediate permeability values. Their structural features, such as embedded halite grains and tightly closed boundaries, suggest limited but present pathways for gas migration.

Table 3—The measured porosity, absolute H₂ permeability considering Klinkenberg effect, and the evaluated Klinkenberg constant.

Sample	ϕ (%)	k_{∞} (m ²)
1	0.73	7.22×10^{-22}
2	1.5	$17.3 \times 10^{-21} \text{ m}^2$
3	1.45	9.21×10^{-21}
4	2.12	2.31×10^{-21}

Conclusions

In this study, we investigated the permeability characteristics of four salt-rock samples using a modified pressure pulse system to conduct hydrogen permeability experiments. The samples included one from the caprock formation (S1), one from the salt layer predominantly composed of crystallized halite (S2), and two from the marlstone interbedded layer (S3, S4). The permeability of the plugs was measured through the pressure-pulse decay method. For each sample, we conducted a series of experiments with varying upstream and downstream pressures to make the average H₂ pressure range from 1 MPa to 7 MPa. For samples S1, S3, and S4, a clear linear relationship was observed between the measured permeability and the reciprocal of average pressure, consistent with the Klinkenberg effect. After correcting for gas slippage, the intrinsic permeabilities were determined to be $7.22 \times 10^{-22} \text{ m}^2$, $9.21 \times 10^{-21} \text{ m}^2$ and $2.31 \times 10^{-21} \text{ m}^2$ respectively. Sample S2, however, deviated from this linear trend, particularly with lower pressures. This behavior is attributed to the influence of Knudsen flow, as hydrogen moves through the narrow, nano-scale pathways within the halite crystals. In addition, there are more grain boundaries in S2 compared to other samples, which are likely to serve as preferential pathways and contribute to its higher intrinsic permeability ($17.3 \times 10^{-21} \text{ m}^2$). Together, these factors highlight the dual role of crystal-scale transport mechanisms and microstructural features in governing hydrogen migration in salt formations, especially in halite-dominated zones.

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